



Error analysis of a predictor-corrector compact difference scheme on nonuniform meshes for the time-fractional BBM equations

Wei Gou¹ · Maohua Ran¹ · Haodong Pu¹

Received: 4 September 2025 / Revised: 3 March 2026 / Accepted: 21 March 2026 /

Published online: 2 April 2026

© The Author(s) under exclusive licence to Korean Society for Informatics and Computational Applied Mathematics 2026

Abstract

This work develops a novel high-order numerical scheme based on non-uniform meshes for the nonlinear time-fractional Benjamin-Bona-Mahony equations. The scheme achieves fourth-order spatial accuracy and second-order temporal accuracy, using a predictor–corrector strategy and a method of order reduction. Two compact difference operators are employed to approximate the nonlinear advection term uu_x and the linear dispersive term u_{xx} . Using the discrete energy method, we establish rigorous error estimates for the scheme in the H^1 -norm. To enhance computational efficiency, exponential sum approximations are utilized, and a fast algorithm is constructed and analyzed. The validity of the proposed schemes is demonstrated, and their performance is compared using two numerical examples.

Keywords Fractional benjamin-bona-mahony equations · Compact difference method · Error estimates · Graded mesh · Predictor-Corrector

MSC Classification 65M06 · 65M12 · 65M15

✉ Maohua Ran
maohuaran@163.com

Wei Gou
gouweig@outlook.com

Haodong Pu
puhaodong10@163.com

¹ School of Mathematical Sciences, Sichuan Normal University, Chengdu, Sichuan 610066, China

1 Introduction

The Benjamin–Bona–Mahony (BBM) equation, introduced by Benjamin et al. [1] to model small-amplitude long waves in nonlinear dispersive media, has been extensively studied for its qualitative properties and numerical solutions. Zhang et al. [2] developed an efficient three-point compact finite difference scheme for the Benjamin–Bona–Mahony–Burgers equation. The scheme achieves fourth-order spatial accuracy and second-order temporal accuracy, with optimal convergence order $O(\tau^2 + h^4)$ in the maximum norm, while requiring only one linear system to be solved at each time level. In a related study, Zhang et al. [3] proposed and analyzed two linearized finite difference schemes for the same equation. These schemes are second-order convergent in both time and space, and are shown to preserve conservation properties, ensure boundedness, and guarantee unique solvability. Moreover, the approach is extended to handle nonlinear source terms via Newton linearization. Further research on the BBM equation can be found in [4–7]. Recent advancements have focused on fractional generalizations of the BBM equation to address complex physical and engineering systems [8–12]. Compared with the classical integer-order BBM equation, the fractional-order BBM equation leverages the nonlocal nature of fractional derivatives to more accurately simulate physical processes involving persistent memory or history dependence. This allows for a more realistic characterization of dissipation, viscosity, and memory effects in wave propagation within complex media, thereby offering distinct advantages in applications such as nonlinear optics and complex fluid dynamics [13, 14]. Various fractional BBM-type equations have been discussed; for instance, Lyu et al. [15] proposed a linearized second-order finite difference scheme for time-fractional generalized BBM equations, achieving unconditional convergence with second-order temporal-spatial accuracy in maximum-norm estimates, Li [16] constructed second-order stable linearized difference schemes for BBM equations with fractional nonlocal viscous terms. Yi et al. [17] proposed and analyzed a high-order implicit–explicit compact finite difference method for the nonlinear time-fractional Benjamin–Bona–Mahony–Burgers equation. The method achieves fourth-order spatial convergence and up to $(3 + \alpha)$ -order temporal convergence for weakly singular solutions, while maintaining simplicity in implementation. Lin et al. [18] developed an iterative radial basis function (RBF)-based method combined with quasi-linearization to solve nonlinear time-fractional Benjamin–Bona–Mahony–Burgers equations in arbitrary domains. Numerical experiments in one, two, and three spatial dimensions demonstrate the high accuracy and efficiency of their approach. In addition, Dhayal et al. [19–21] have investigated the stochastic Benjamin–Bona–Mahony equation.

In this work, we develop and analyze a high-order compact difference scheme for the following time-fractional BBM initial-boundary value problem:

$$\begin{cases} {}^C D_0^\alpha (u - u_{xx}) + \gamma uu_x - \lambda u_{xx} = f(x, t, u), & (x, t) \in \Omega \times (0, T], \\ u(x, 0) = \varphi(x), & x \in \bar{\Omega} \\ u(x, t) = 0, & u(x, t) \in \partial\Omega \times (0, T] \end{cases} \quad (1)$$

where $\Omega = (x_L, x_R)$, with $\lambda(\geq 0)$ and γ are constants, and f denotes a forcing term satisfying the Lipschitz condition:

$$|f(x, t, u) - f(x, t, v)| \leq L|u - v|, \quad (x, t) \in \bar{\Omega} \times [0, T], \quad (2)$$

where the Lipschitz constant $L \geq 0$, $u, v \in \mathbb{R}$, and the Caputo fractional derivative of order $\alpha \in (0, 1)$ is defined as:

$${}^C D_0^\alpha w(t) = \frac{1}{\Gamma(1-\alpha)} \int_{s=0}^t (t-s)^{-\alpha} w'(s) ds. \quad (3)$$

We assume the problem (1) admits a unique solution satisfying the regularity conditions:

$$\left| \frac{\partial^l u}{\partial x^l}(x, t) \right| \leq C, \quad l = 3, 4, \quad (4)$$

$$\left| \frac{\partial^{l+k} u}{\partial x^l \partial t^k}(x, t) \right| \leq C(1 + t^{\alpha-k}), \quad l, k = 0, 1, 2, \quad (5)$$

for $(x, t) \in \bar{\Omega} \times (0, T]$ with constant $C > 0$. These regularity assumptions are justified by the characteristic singularity t^α near $t \rightarrow 0^+$ in solutions of Caputo-type time-fractional differential equations, see [22–25]. The bounds in (4)–(5) indicate that u exhibits weak initial singularity at $t=0$. While existing works mentioned earlier [15, 16] implicitly assume sufficient solution regularity, Lyu [22] introduced a temporal nonuniform mesh technique using high-order interpolation and central differences, and Wang [26] proposed a linearized compact difference method, Zhou [27] generalised the Adams-type predictor-corrector method of [28] to graded meshes and developed a fast second-order predictor-corrector method for the nonlinear time-fractional BBM-Burgers equation. However, the approximations for the non-linear advection term uu_x proposed by Zhou [27] and Lyu [22] are confined to second-order accuracy in space, which inherently limits their applicability in large-scale simulations. Most of the above-mentioned studies either suffer from low accuracy or involve high computational complexity and significant time consumption. Based on the above analysis, we present a novel compact difference method.

The main contributions of this paper are summarized as follows:

- A novel high-order compact predictor–corrector scheme is proposed for nonlinear time-fractional BBM equations on non-uniform meshes;
- Rigorous H^1 -norm error estimates are established using the discrete energy method;
- A fast implementation is developed through exponential sum approximation to significantly improve computational efficiency.

The remainder of this paper is structured as follows: Section 2 develops the numerical scheme through transformation to a Volterra integral equation. Section 3 establishes convergence via a new Gronwall-type inequality. Section 4 presents numerical verification, with conclusions in Section 5.

Throughout this work, we employ the convention that C (with or without subscripts) denotes a generic positive constant that may vary in value at different occurrences, but remains independent of spatial and temporal discretization sizes.

Remark 1.1 It is worth emphasizing that the proposed scheme employs a different compact treatment of the nonlinear advection term uu_x , as proposed by Wang et al. [29], combined with an order reduction strategy for u_{xx} . This design results in a compact stencil that requires fewer boundary points while still achieving fourth-order spatial accuracy and preserving the temporal convergence performance.

2 Construction of the compact difference schemes

2.1 Predictor-corrector compact difference scheme

Let $v = u_{xx}$. Applying the Riemann-Liouville fractional integral of order α to both sides of (1) yields

$$\begin{cases} u(x, t) - v(x, t) = G(x) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} F(x, s, u) ds, \\ v(x, t) = u_{xx}, \quad (x, t) \in \Omega \times (0, T], \\ u(x, 0) = \varphi(x), \quad x \in \bar{\Omega}, \\ u(x, t) = 0, \quad u(x, t) \in \partial\Omega \times (0, T], \end{cases} \quad (6)$$

where

$$\begin{aligned} G(x) &= u(x, 0) - v(x, 0) = \varphi(x) - (\varphi_{xx})(x), \\ F(x, t, u) &= \lambda v(x, t) - \gamma(uu_x)(x, t) + f(x, t, u(x, t)). \end{aligned}$$

At t_{n+1} , the first equation in (6) becomes

$$u(x, t_{n+1}) - v(x, t_{n+1}) = G(x) + G_h(x, t_{n+1}) + G_l(x, t_{n+1}), \quad (7)$$

where the historical and local terms

$$\begin{aligned} G_h(x, t_{n+1}) &= \frac{1}{\Gamma(\alpha)} \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} (t_{n+1} - s)^{\alpha-1} F(x, s, u) ds, \\ G_l(x, t_{n+1}) &= \frac{1}{\Gamma(\alpha)} \int_{t_n}^{t_{n+1}} (t_{n+1} - s)^{\alpha-1} F(x, s, u) ds. \end{aligned}$$

Define the spatial and temporal discretizations

$$h = (x_R - x_L)/M, \quad x_i = x_L + ih, \quad \text{for } 0 \leq i \leq M, \\ t_n = T(n/N)^r, \quad \tau_n = t_n - t_{n-1}, \quad \text{for } 1 \leq n \leq N,$$

here, $r \geq 1$ is a user-chosen mesh-grading parameter, with index sets

$$\Omega_h = \{1, \dots, M-1\}, \quad \Omega_\tau = \{1, \dots, N-1\}, \quad \Gamma_h = \{0, M\},$$

where M and N are positive integers. For any mesh function $\omega \in \{\omega_i^n | (i, n) \in \Omega_h \times \Omega_\tau\}$, define

$$\delta_x \omega_i = \frac{\omega_i - \omega_{i-1}}{h}, \quad \Delta_x \omega_i = \frac{\omega_{i+1} - \omega_{i-1}}{2h}, \quad \delta_x^2 \omega_i = \frac{\omega_{i+1} - 2\omega_i + \omega_{i-1}}{h^2}, \\ (\mathcal{A}\omega_i^n) = \frac{1}{12} \left(\omega_{i+1}^n + 10\omega_i^n + \omega_{i-1}^n \right) = \left(\mathcal{I} + \frac{h^2}{12} \delta_x^2 \right) \omega_i^n.$$

Lemma 2.1 ([6]) Suppose $w(x) \in C_x^6(\Omega)$. Then it holds that

$$\frac{1}{12} \left[w''(x_{i+1}) + 10w''(x_i) + w''(x_{i-1}) \right] \\ = \frac{1}{h^2} \left[w(x_{i+1}) - 2w(x_i) + w(x_{i-1}) \right] + \frac{h^4}{240} w^{(6)}(\xi_i), \quad i \in \Omega_h,$$

where $\xi_i \in (x_{i-1}, x_{i+1})$.

Lemma 2.2 ([29]) For $u \in \mathcal{V}_h = \{v | v = (v_0, v_1, \dots, v_M)\}$ with $A(x) = u_{xx}(x)$. If $u(x) \in C^5([x_{i-1}, x_{i+1}])$, we have

$$u(x_i)u_x(x_i) = \varphi(A, u)_i + O(h^4), \quad i \in \Omega_h,$$

where

$$\varphi(v, u)_i = \psi(u, u)_i - \frac{h^2}{2} \psi(v, u)_i, \quad \psi(u, v)_i = \frac{1}{3} [u_i \Delta_x v_i + \Delta_x (uv)_i].$$

Let $U_i^n = u(x_i, t_n)$ and $V_i^n = v(x_i, t_n)$. Using the compact operator $\mathcal{A}$ and Lemma 2.1, gives

$$\mathcal{A}v(x_i, t_n) = \mathcal{A}u_{xx}(x_i, t_n) = \left(\mathcal{I} + \frac{h^2}{12} \delta_x^2 \right) V_i^n = \delta_x^2 U_i^n + \frac{h^4}{240} \frac{\partial^6 u(\xi_i, t_n)}{\partial x^6},$$

where $\xi_i \in (x_{i-1}, x_{i+1})$. Consequently, we have

$$V_i^n = \delta_x^2 U_i^n - \frac{h^2}{12} \delta_x^2 V_i^n + \frac{h^4}{240} \frac{\partial^6 u(\xi_i, t_n)}{\partial x^6} = \delta_x^2 U_i^n - \frac{h^2}{12} \delta_x^2 V_i^n + O(h^4).$$

Moreover, from Lemma 2.2, there exists an approximation for the nonlinear term uu_x as follows

$$u(x_i, t_n)u_x(x_i, t_n) = \varphi(V^n, U^n)_i + O(h^4). \quad (8)$$

In the construction of the proposed scheme, linear interpolation is utilized to approximate the function $F(x, t, u)$ within the fractional integral. Specifically, for $1 \leq j \leq n$, $1 \leq i \leq M-1$, the linear interpolation operator $\Pi_j F(x_i, t, u)$ of $F(x_i, t, u)$ over the interval $[t_j, t_{j+1}]$ is defined as

$$\Pi_j F(x_i, t, u) = \frac{t - t_{j+1}}{t_j - t_{j+1}} F_i^j + \frac{t - t_j}{t_{j+1} - t_j} F_i^{j+1} = L_{j,0}(t) F_i^j + L_{j,1}(t) F_i^{j+1},$$

where $F_i^j = \lambda v_i^j - \gamma \varphi(v^j, u^j)_i + f(x_i, t_j, u_i^j)$ and $t \in [t_j, t_{j+1}]$.

The predictor-corrector procedure, adopting the framework in [27], comprises two stages:

Stage 1. Prediction Phase:

- For $n = 0$: the term $u(x_i, t_1) - v(x_i, t_1) = G(x_i) + G_l(x_i, t_1)$ is approximated by replacing $F(x_i, s, u)$ in $G_l(x_i, t_1)$ with F_i^0 . This yields the predicted solutions $(u^p)_i^1$ and $(v^p)_i^1$ for $u(x_i, t_1)$ and $v(x_i, t_1)$, respectively.
- For $n \geq 1$: over each interval $[t_j, t_{j+1}]$ ($j = 0, 1, \dots, n-1$), $F(x_i, s, u)$ is replaced by the linear interpolation of F_i^j and F_i^{j+1} . On the final interval $[t_n, t_{n+1}]$, $F(x_i, s, u)$ is substituted with the linear function interpolating F_i^{n-1} and F_i^n . This construction yields the approximation $(u^p)_i^{n+1}$ and $(v^p)_i^{n+1}$ for $u(x_i, t_{n+1})$ and $v(x_i, t_{n+1})$, respectively.

Stage 2. Correction Phase:

The approximation of $G_l(x, t_{n+1})$ follows the prediction phase method, except on $[t_n, t_{n+1}]$, where $F(x_i, s, u)$ is replaced by the linear interpolant of F_i^n and $(F^p)_i^{n+1}$, where $(F^p)_i^{n+1} = \lambda (v^p)_i^{n+1} - \gamma \varphi((v^p)^{n+1}, (u^p)^{n+1})_i + f(x_i, t_{n+1}, (u^p)_i^{n+1})$.

According to the above, we can derive the predictor-corrector compact difference (PCCD) scheme as follows:

$$\left\{ \begin{array}{l} (u^p)_i^{n+1} - (v^p)_i^{n+1} \\ = G(x_i) + \frac{1}{\Gamma(\alpha)} \sum_{j=0}^{n-1} (a_{j,0}^{n+1} F_i^j + a_{j,1}^{n+1} F_i^{j+1}) + \frac{1}{\Gamma(\alpha)} (b_{n-1,0}^{n+1} F_i^{n-1} + b_{n-1,1}^{n+1} F_i^n), \\ u_i^{n+1} - v_i^{n+1} \\ = G(x_i) + \frac{1}{\Gamma(\alpha)} \sum_{j=0}^{n-1} (a_{j,0}^{n+1} F_i^j + a_{j,1}^{n+1} F_i^{j+1}) + \frac{1}{\Gamma(\alpha)} [a_{n,0}^{n+1} F_i^n + a_{n,1}^{n+1} (F^p)_i^{n+1}], \\ (v^p)_i^{n+1} = \delta_x^2 (u^p)_i^{n+1} - \frac{h^2}{12} \delta_x^2 (v^p)_i^{n+1}, \quad (i, n) \in \Omega_h \times \Omega_\tau, \\ v_i^{n+1} = \delta_x^2 u_i^{n+1} - \frac{h^2}{12} \delta_x^2 v_i^{n+1}, \quad (i, n) \in \Omega_h \times \Omega_\tau, \\ F_i^j = \lambda v_i^j - \gamma \varphi(v^j, u^j)_i + f(x_i, t_j, u_i^j), \\ (F^p)_i^{n+1} = \lambda (v^p)_i^{n+1} - \gamma \varphi((v^p)^{n+1}, (u^p)^{n+1})_i + f(x_i, t_{n+1}, (u^p)_i^{n+1}), \end{array} \right.$$

where $F^{-1} = 0$, $b_{-1,0}^1 = 0$, $b_{-1,1}^1 = \tau_1^\alpha / \alpha$, and

$$\begin{aligned} a_{j,0}^{n+1} &= \int_{t_j}^{t_{j+1}} (t_{n+1} - s)^{\alpha-1} L_{j,0}(s) ds \\ &= \frac{1}{\tau_{j+1}} \left[\frac{(t_{n+1} - t_j)^{\alpha+1} - (t_{n+1} - t_{j+1})^{\alpha+1}}{\alpha + 1} - (t_{n+1} - t_{j+1}) \frac{(t_{n+1} - t_j)^\alpha - (t_{n+1} - t_{j+1})^\alpha}{\alpha} \right], \\ a_{j,1}^{n+1} &= \int_{t_j}^{t_{j+1}} (t_{n+1} - s)^{\alpha-1} L_{j,1}(s) ds \\ &= \frac{1}{\tau_{j+1}} \left[(t_{n+1} - t_j) \frac{(t_{n+1} - t_j)^\alpha - (t_{n+1} - t_{j+1})^\alpha}{\alpha} - \frac{(t_{n+1} - t_j)^{\alpha+1} - (t_{n+1} - t_{j+1})^{\alpha+1}}{\alpha + 1} \right], \\ a_{n,0}^{n+1} &= \int_{t_n}^{t_{n+1}} (t_{n+1} - s)^{\alpha-1} L_{n,0}(s) ds = \frac{\tau_{n+1}^\alpha}{(\alpha + 1)}, \\ a_{n,1}^{n+1} &= \int_{t_n}^{t_{n+1}} (t_{n+1} - s)^{\alpha-1} L_{n,1}(s) ds = \frac{\tau_{n+1}^\alpha}{\alpha(\alpha + 1)}, \\ b_{n-1,0}^{n+1} &= \int_{t_n}^{t_{n+1}} (t_{n+1} - s)^{\alpha-1} L_{n-1,0}(s) ds = -\frac{\tau_{n+1}^{\alpha+1}}{\tau_n(\alpha(\alpha + 1))}, \\ b_{n-1,1}^{n+1} &= \int_{t_n}^{t_{n+1}} (t_{n+1} - s)^{\alpha-1} L_{n-1,1}(s) ds = \frac{\tau_{n+1}^\alpha}{\tau_n} \left(\frac{\tau_{n+1}}{\alpha(\alpha + 1)} + \frac{\tau_n}{\alpha} \right). \end{aligned}$$

2.2 Fast predictor-corrector compact difference scheme

To reduce the computational cost of the above PCCD scheme, we adopt the sum-of-exponentials approximation proposed in [30], formulated as the following lemma.

Lemma 2.3 ([30]) Given $\beta \in (0, 2)$, an absolute tolerance error ε , a cut off time $\Delta t = \min_{1 \leq n \leq N} \tau_n$ and a final time T , there exists a positive integer N_{exp} , quadrature nodes s_i and corresponding positive weights ϖ_l such that

$$\left| t^{-\beta} - \sum_{l=1}^{N_{exp}} \varpi_l e^{-s_l t} \right| \leq \varepsilon \quad \text{for } t \in [\Delta t, T],$$

where

$$N_{exp} = O \left(\left(\log \frac{1}{\varepsilon} \right) \left(\log \log \frac{1}{\varepsilon} + \log \frac{T}{\Delta t} \right) + \left(\log \frac{1}{\Delta t} \right) \left(\log \log \frac{1}{\varepsilon} + \log \frac{1}{\Delta t} \right) \right).$$

Inspired by the aforementioned lemma, we now develop the fast predictor-corrector scheme for the equivalent problem (6). First, to avoid confusion, we introduce the discrete solution components of the fast method:

$$\bar{U}_i^{n+1} = u(x_i, t_{n+1}), \quad (\bar{U}^p)_i^{n+1} = (u^p)(x_i, t_{n+1}), \quad \bar{F}^j = F(x, t_j, u(x_i, t_j)),$$

with $\bar{V}$, $\bar{V}^p$ defined analogously. The numerical approximation combines two key components:

- The local term $G_l(x, t_{n+1})$ is handled identically to the standard PCCD scheme;
- For the historical part $G_h(x, t_{n+1})$, we: Replace the kernel $(t_{n+1} - s)^{\alpha-1}$ with its SOE approximation;
- Approximate $F(x, s, u)$ using linear interpolation between $\bar{F}^j$ and $\bar{F}^{j+1}$.

Following [27], combined with Lemma 2.2, we can from (7) derive a complete fast predictor-corrector compact difference (FPCCD) scheme as follows:

$$\begin{cases} (\bar{u}^P)_i^{n+1} - (\bar{v}^P)_i^{n+1} = G(x_i) + \frac{1}{\Gamma(\alpha)} \left(\sum_{l=1}^{N_{exp}} \varpi_l Q_i^n + b_{n-1,0}^{n+1} \bar{F}_i^{n-1} + b_{n-1,1}^{n+1} \bar{F}_i^n \right), \\ \bar{u}_i^{n+1} - \bar{v}_i^{n+1} = G(x_i) + \frac{1}{\Gamma(\alpha)} \left(\sum_{l=1}^{N_{exp}} \varpi_l Q_i^n + a_{n,0}^{n+1} \bar{F}_i^n + a_{n,1}^{n+1} (\bar{F}^p)_i^{n+1} \right), \\ Q_i^0 = 0, \\ Q_i^n = e^{-s_l \tau_{n+1}} Q_i^{n-1} + A_{i,0}^{n+1} \bar{F}_i^{n-1} + A_{i,1}^{n+1} \bar{F}_i^n, \text{ for } 1 \leq i \leq N_{exp}, \quad 1 \leq n \leq N-1, \\ \bar{u}_i^0 = \varphi(x_i), \quad \text{for } 0 \leq i \leq M, \quad \bar{u}_0^n = \bar{u}_M^n = 0, \quad \text{for } 0 < n \leq N, \end{cases}$$

where

$$\begin{cases} \bar{v}_i^{n+1} = \delta_x^2 \bar{u}_i^{n+1} - \frac{h^2}{12} \delta_x^2 \bar{v}_i^{n+1}, & (i, n) \in \Omega_h \times \Omega_\tau, \\ (\bar{v}^p)_i^{n+1} = \delta_x^2 (\bar{u}^p)_i^{n+1} - \frac{h^2}{12} \delta_x^2 (\bar{v}^p)_i^{n+1}, & (i, n) \in \Omega_h \times \Omega_\tau, \\ \bar{F}_i^j = \lambda \bar{v}_i^j - \gamma \varphi(\bar{v}^j, \bar{u}^j)_i + f(x_i, t_{n+1}, \bar{u}_i^j), \\ (\bar{F}^p)_i^{n+1} = \lambda (\bar{v}^p)_i^{n+1} - \gamma \varphi((\bar{v}^p)^{n+1}, (\bar{u}^p)^{n+1})_i + f(x_i, t_{n+1}, (\bar{u}^p)_i^{n+1}), \end{cases}$$

and the integration coefficients

$$\begin{aligned} A_{i,0}^{n+1} &= \int_{s=t_{n-1}}^{t_n} e^{-s_l(t_{n+1}-s)} L_{n-1,0}(s) ds = \frac{e^{-s_l \tau_{n+1}}}{\tau_n s_l^2} [(\tau_n s_l + 1) e^{-s_l \tau_n} - 1], \\ A_{i,1}^{n+1} &= \int_{s=t_{n-1}}^{t_n} e^{-s_l(t_{n+1}-s)} L_{n-1,1}(s) ds = \frac{1}{\tau_n s_l^2} [e^{-s_l \tau_{n+1}} - e^{-s_l(\tau_n + \tau_{n+1})}]. \end{aligned}$$

3 High-order convergence analysis

Define the discrete function spaces:

$$\mathcal{V}_h = \{v | v = (v_0, v_1, \dots, v_M)\}, \quad \mathcal{V}_h^\circ = \{v | v \in \mathcal{V}_h, v_0 = v_M = 0\}.$$

For $u, v \in \mathcal{V}_h^\circ$, we introduce the following inner products and norms:

$$\langle u, v \rangle = h \sum_{i=1}^{M-1} u_i v_i, \quad \|v\| = \sqrt{\langle v, v \rangle}, \quad |v|_1 = \sqrt{\langle \delta_x v, \delta_x v \rangle}, \quad \|v\|_{H^1} = \sqrt{\|v\|^2 + |v|_1^2}.$$

3.1 Auxiliary lemmas

Lemma 3.1 ([31]) For any mesh function $v \in \mathcal{V}_h^\circ$, the following inequalities hold:

$$\|v\| \leq \frac{L}{\sqrt{6}} |v|_1, \quad \|v\|_\infty \leq \frac{\sqrt{L}}{2} |v|_1, \quad |v|_1 \leq \frac{2}{h} \|v\|.$$

Lemma 3.2 ([26]) For any mesh functions $u, v \in \mathcal{V}_h^\circ$, we have

$$\langle \delta_x^2 u, u \rangle = -|u|_1^2, \quad |\langle \delta_x^2 u, v \rangle| \leq |u|_1 |v|_1,$$

and

$$\langle \mathcal{A}u, u \rangle \geq \frac{2}{3} \|u\|^2, \quad \|\mathcal{A}u\| \leq \|u\|.$$

Lemma 3.3 ([27]) The coefficients $a_{j,0}^{n+1}, a_{j,1}^{n+1}, b_{n-1,0}^{n+1}, b_{n-1,1}^{n+1}$ satisfy the following estimates:

- $0 < a_{j,\theta}^{n+1} \leq C\tau_{j+1}(t_{n+1} - t_j)^{\alpha-1}$ for $\theta = 0, 1$ and $0 \leq j \leq n$,
- $|b_{n-1,0}^{n+1}| \leq C\tau_n(t_{n+1} - t_{n-1})^{\alpha-1}$, $0 < b_{n-1,1}^{n+1} \leq C\tau_{n+1}^\alpha$, $1 \leq n \leq N-1$,
- $\sum_{j=0}^{n-1} (a_{j,0}^{n+1} + a_{j,1}^{n+1}) + |b_{n-1,0}^{n+1}| + b_{n-1,1}^{n+1} \leq CT^\alpha$, $1 \leq n \leq N-1$,

where C is some positive constant.

Lemma 3.4 ([27]) Let sequences $\{\phi_n\}_{n=0}^N$ and $\{\phi_n^P\}_{n=0}^N$ satisfy

$$\begin{cases} 0 \leq \phi_0 \leq d_0, \\ 0 \leq \phi_{n+1}^P \leq d_0 + \sum_{j=0}^n (k_{1,j}\phi_j + k_{2,j}\phi_j^2), \quad 0 \leq n \leq N-1, \\ 0 \leq \phi_{n+1} \leq d_0 + \sum_{j=0}^n (k_{3,j}\phi_j + k_{4,j}\phi_j^2) + C_2(\phi_{n+1}^P + (\phi_{n+1}^P)^2), \quad 0 \leq n \leq N-1, \end{cases}$$

where C_2 is a fixed constant and $\{k_{\theta,j}\}_{j=0}^N$, $(\theta = 1, 2, 3, 4)$ satisfy

$$k_{\theta,j} = C_2\tau_{j+1}(t_{n+1} - t_j)^{\alpha-1}, \quad 0 \leq j \leq n, \quad 0 \leq n \leq N-1.$$

Define:

$$\begin{aligned} \bar{C}_0 &= \max \{1 + k_{3,0} + k_{4,0}d_0 + C_2(1 + k_{1,0} + k_{2,0}d_0) + C_2d_0(1 + k_{1,0} + k_{2,0}d_0)^2, C_p\}, \\ \hat{C}_0 &= 1 + \bar{C}_0 \sum_{j=0}^n (k_{1,j} + k_{2,j}), \end{aligned}$$

where C_p is an N -independent constant dependent on C_2 . If $\bar{C}_0d_0 \leq 1$ and

$$\hat{C}_0d_0 \leq 1, \quad 0 \leq n \leq N-1,$$

then

$$\phi_{n+1} \leq \bar{C}_0d_0, \quad 0 \leq n \leq N-1. \quad (9)$$

3.2 Error analysis

Define the error functions:

$$e_i^n = U_i^n - u_i^n, \quad (e^p)_i^n = U_i^n - (u^p)_i^n, \quad \xi_i^n = V_i^n - v_i^n, \quad (\xi^p)_i^n = V_i^n - (v^p)_i^n,$$

with vector representations:

$$e^n = (e_1^n, e_2^n, \dots, e_{M-1}^n)^T, \quad (e^p)^n = ((e^p)_1^n, (e^p)_2^n, \dots, (e^p)_{M-1}^n)^T, \\ \xi^n = (\xi_1^n, \xi_2^n, \dots, \xi_{M-1}^n)^T, \quad (\xi^p)^n = ((\xi^p)_1^n, (\xi^p)_2^n, \dots, (\xi^p)_{M-1}^n)^T.$$

Theorem 3.1 Let u be the exact solution of problem (1), satisfying the regularity conditions (4) and (5). Let $\{u_i^n\}$ denote the numerical solution of the PCCD scheme with grading parameter $r = 3/2\alpha$. Then, for sufficiently large N and sufficiently small h , there exists a constant C such that:

$$\|U^n - u^n\|_{H^1} \leq C(N^{-2} + h^4), \quad 0 \leq n \leq N.$$

Proof From the derivation of the PCCD scheme, we can obtain the error equation:

$$e_i^{n+1} - \xi_i^{n+1} \\ = \frac{1}{\Gamma(\alpha)} \left[\sum_{j=0}^{n-1} (a_{j,0}^{n+1} \omega_i^j + a_{j,1}^{n+1} \omega_i^{j+1}) + a_{n,0}^{n+1} \omega_i^n + a_{n,1}^{n+1} (\omega^p)_i^{n+1} \right] \\ + (r_1)_i^{n+1} + (r_2)_i^{n+1} + (r_3)_i^{n+1}, \quad (10)$$

and

$$(e^p)_i^{n+1} - (\xi^p)_i^{n+1} \\ = \frac{1}{\Gamma(\alpha)} \left[\sum_{j=0}^{n-1} (a_{j,0}^{n+1} \omega_i^j + a_{j,1}^{n+1} \omega_i^{j+1}) + b_{n-1,0}^{n+1} \omega_i^{n-1} + b_{n-1,1}^{n+1} \omega_i^n \right] \\ + (r_1)_i^{n+1} + (r_4)_i^{n+1} + (r_5)_i^{n+1}, \quad (11)$$

where

$$\omega_i^j = \lambda \xi_i^j - \gamma [\varphi(V_i^j, U_i^j) - \varphi(v_i^j, u_i^j)] + f(x_i, t_j, U_i^j) - f(x_i, t_j, u_i^j), \quad (12)$$

$$(\omega^p)_i^{n+1} = \lambda (\xi^p)_i^{n+1} - \gamma [\varphi(V_i^{n+1}, U_i^{n+1}) - \varphi((v^p)_i^{n+1}, (u^p)_i^{n+1})] \\ + f(x_i, t_{n+1}, U_i^{n+1}) - f(x_i, t_{n+1}, (u^p)_i^{n+1}), \quad (13)$$

$$\text{and } \xi_i^j = \delta_x^2 e_i^j - \frac{h^2}{12} \delta_x^2 \xi_i^j,$$

$$\left\{ \begin{array}{l} (r_1)_i^{n+1} = u_{xx}(x_i, t_{n+1}) - \left(\delta_x^2 U_i^n - \frac{h^2}{12} \delta_x^2 V_i^n \right) \\ (r_2)_i^{n+1} = \frac{1}{\Gamma(\alpha)} \sum_{j=0}^n \left(a_{j,0}^{n+1} (r_u)_i^j + a_{j,1}^{n+1} (r_u)_i^{j+1} \right), \\ (r_3)_i^{n+1} = \frac{1}{\Gamma(\alpha)} \sum_{j=0}^n \int_{s=t_j}^{t_{j+1}} (t_{n+1} - s)^{\alpha-1} (F - \Pi_j F)(x_i, s, u) ds, \\ (r_4)_i^{n+1} = \frac{1}{\Gamma(\alpha)} \left[\sum_{j=0}^{n-1} \left(a_{j,0}^{n+1} (r_u)_i^j + a_{j,1}^{n+1} (r_u)_i^{j+1} \right) + b_{n-1,0}^{n+1} (r_u)_i^{n-1} + b_{n-1,1}^{n+1} (r_u)_i^n \right], \\ (r_5)_i^{n+1} = \frac{1}{\Gamma(\alpha)} \left[\sum_{j=0}^{n-1} \int_{s=t_j}^{t_{j+1}} (t_{n+1} - s)^{\alpha-1} (F - \Pi_j F)(x_i, s, u) ds \right. \\ \left. + \int_{s=t_n}^{t_{n+1}} (t_{n+1} - s)^{\alpha-1} (F - \Pi_{n-1} F)(x_i, s, u) ds \right], \end{array} \right.$$

with

$$(r_u)_i^j = \lambda \left[u_{xx}(x_i, t_j) - \left(\delta_x^2 U_i^j - \frac{h^2}{12} \delta_x^2 V_i^j \right) \right] - \gamma [(uu_x)(x_i, t_j) - \varphi(V^j, U^j)]_i.$$

Using the regularity assumption (4) and Lemmas 2.1 and 2.2, we can find that

$$|(r_1)_i^{n+1}| + |(r_2)_i^{n+1}| + |(r_4)_i^{n+1}| \leq Ch^4.$$

From [27](Theorem 2.1), when grading parameter $r = \frac{3}{2\alpha}$, we can obtain:

$$|(r_3)_i^{n+1}| + |(r_5)_i^{n+1}| \leq CN^{-2}.$$

Taking the inner product of (12) with e^{n+1} gives:

$$\begin{aligned} & \langle \omega^j, e^{n+1} \rangle \\ &= \lambda \langle \xi^j, e^{n+1} \rangle - \gamma \langle (\varphi(V^j, U^j) - \varphi(v^j, u^j)), e^{n+1} \rangle + \langle f(\cdot, t_j, U^j) - f(\cdot, t_j, u^j), e^{n+1} \rangle. \end{aligned} \quad (14)$$

Applying the Cauchy-Schwarz inequality and Lemma 3.2, we derive:

$$\lambda \langle \xi^j, e^{n+1} \rangle = \lambda \langle \delta_x^2 e^j - \frac{h^2}{12} \delta_x^2 \xi^j, e^{n+1} \rangle \leq \lambda C (|e^j|_1 |e^{n+1}|_1 + \frac{h^2}{12} |\xi^j|_1 |e^{n+1}|_1). \quad (15)$$

Noticing that the Lipschitz condition of f yields:

$$|\langle f(\cdot, t_j, U^j) - f(\cdot, t_j, u^j), e^{n+1} \rangle| \leq LC \|e^j\| \cdot \|e^{n+1}\|. \quad (16)$$

Based on the above similar analysis and imply (4) for the second term on the right in (14) ignores γ gives:

$$\begin{aligned}
& -\langle \varphi(V^j, U^j) - \varphi(v^j, u^j), e^{n+1} \rangle \\
& = -\langle \psi(e^j, U^j), e^{n+1} \rangle + \frac{h^2}{2} \langle \psi(\xi^j, U^j), e^{n+1} \rangle \\
& = -\frac{h}{3} \sum_{i=1}^{M-1} \left[e_i^j \Delta_x U_i^j + \Delta_x (e_i^j U_i^j) \right] e_i^{n+1} + \frac{h^3}{6} \sum_{i=1}^{M-1} \left[\xi_i^j \Delta_x U_i^j + \Delta_x (\xi_i^j U_i^j) \right] e_i^{n+1} \quad (17) \\
& = -\frac{h}{3} \sum_{i=1}^{M-1} \left[e_i^j \Delta_x U_i^j + \frac{1}{2} e_{i+1}^j \delta_x U_{i+\frac{1}{2}}^j \right] e_i^{n+1} + \frac{h^3}{6} \sum_{i=1}^{M-1} \left[\xi_i^j \Delta_x U_i^j + \frac{1}{2} \xi_{i+1}^j \delta_x U_{i+\frac{1}{2}}^j \right] e_i^{n+1} \\
& \leq \frac{C}{2} \|e^j\| \|e^{n+1}\| + \frac{h^2}{4} C \|\xi^j\| \|e^{n+1}\|.
\end{aligned}$$

Combining (15)-(17) in (14) produces:

$$\begin{aligned}
\langle \omega^j, e^{n+1} \rangle & \leq \lambda C \left(|e^j|_1 |e^{n+1}|_1 + \frac{h^2}{12} |\xi^j|_1 |e^{n+1}|_1 \right) + LC \|e^j\| \|e^{n+1}\| \\
& \quad + \frac{C}{2} \gamma \|e^j\| \|e^{n+1}\| + \frac{h^2}{4} \gamma C \|\xi^j\| \|e^{n+1}\| \\
& \leq C (\lambda + L + \gamma) \|e^j\|_{H^1} \|e^{n+1}\|_{H^1} + \frac{h^2}{12} \lambda C \|\xi^j\|_{H^1} \|e^{n+1}\|_{H^1} \quad (18) \\
& \quad + \frac{h^2}{4} \gamma C \|\xi^j\|_{H^1} \|e^{n+1}\|_{H^1} \\
& \leq C \hat{e}^j \|e^{n+1}\|_{H^1},
\end{aligned}$$

where

$$\hat{e}^j = (\lambda + L + \gamma) \|e^j\|_{H^1} + (\lambda + \gamma) \|\xi^j\|_{H^1}.$$

Following analogous procedures, from (2), (5), Lemma 3.2 and (11), we can get:

$$\langle (\omega^p)^{n+1}, e^{n+1} \rangle \leq C (\hat{e}^p)^{n+1} \|e^{n+1}\|_{H^1}, \quad (19)$$

where

$$(\hat{e}^p)^{n+1} = (\lambda + L + \gamma) \|(e^p)^{n+1}\|_{H^1} + (\lambda + \gamma) \|(\xi^p)^{n+1}\|_{H^1}. \quad (20)$$

Multiplying both sides of (10) by $h e_i^{n+1}$, summing over i , and incorporating (18)–(20), from Lemma 3.2 and $\xi_i^{n+1} = \delta_x^2 e_i^{n+1} - \frac{h^2}{12} \delta_x^2 \xi_i^{n+1}$, we can obtain:

$$\begin{aligned}
\|e^{n+1}\|_{H^1}^2 &\leq C\|e^{n+1}\|_{H^1} \left[\sum_{j=0}^{n-1} \left(a_{j,0}^{n-1} \hat{e}_i^j + a_{j,1}^{n+1} \hat{e}_i^{j+1} \right) + a_{n,0}^{n+1} \hat{e}^n + a_{n,1}^{n+1} (\hat{e}^p)^{n+1} \right] \\
&\quad + \frac{h^2}{12} |\xi^{n+1}|_1 |e^{n+1}|_1 + C\|e^{n+1}\|_{H^1} \|(r_1)_i^{n+1} + (r_2)_i^{n+1} + (r_3)_i^{n+1}\| \\
&\leq C\|e^{n+1}\|_{H^1} \left[\sum_{j=1}^n (a_{j-1,1}^{n+1} + a_{j,0}^{n+1}) \hat{e}^j + \tau_{n+1}^\alpha (\hat{e}^p)^{n+1} \right] \\
&\quad + \frac{h^2}{12} |\xi^{n+1}|_1 |e^{n+1}|_1 + C\|e^{n+1}\|_{H^1} (N^{-2} + h^4).
\end{aligned} \tag{21}$$

Invoking Lemma 3.1 and Lemma 3.2 for (21), we deduce:

$$\|e^{n+1}\|_{H^1} \leq C_2 \sum_{j=1}^n (a_{j-1,1}^{n+1} + a_{j,0}^{n+1}) \hat{e}^j + C_2 \tau_{n+1}^\alpha (\hat{e}^p)^{n+1} + C_2 (N^{-2} + h^4). \tag{22}$$

A parallel analysis for (11), from Lemma 3.2, Lemma 3.3, and (18)-(20) yields:

$$\begin{aligned}
\|(e^p)^{n+1}\|_{H^1} &\leq C_3 \sum_{j=1}^{n-1} (a_{j-1,1}^{n+1} + a_{j,0}^{n+1}) (\hat{e})^j + C_3 a_{n-1,1}^{n+1} (\hat{e})^n + C_3 |b_{n-1,0}^{n+1}| (\hat{e})^{n-1} \\
&\quad + C_3 b_{n-1,1}^{n+1} (\hat{e})^n + C_3 (N^{-2} + h^4).
\end{aligned} \tag{23}$$

The remainder follows via induction and application of Lemma 3.4, paralleling the proof of Theorem 3.1 in [27].

Following an analogous argument to Theorem 3.1, we can establish the convergence of the FPCCD scheme.

Theorem 3.2 Let u and $\{\bar{u}\}_i^n$ denote the solutions to the problem (6) and the FPCCD scheme, respectively, with grading parameter $r = \frac{3}{2\alpha}$. Under assumptions (4) and (5), for all sufficiently large N and all sufficiently small h , there exists a constant C such that

$$\|\bar{U}^n - \bar{u}^n\|_{H^1} \leq C\varepsilon + C(N^{-2} + h^4), \quad 1 \leq n \leq N.$$

Remark 3.1 Theorem 3.1 demonstrates the convergence of the proposed compact difference scheme, which achieves second-order accuracy in time and fourth-order accuracy in space. The proof is analogous to that given by Zhou et al. [27], with the key point being the scaling estimate provided in inequality (17). Since the arguments are similar to those used in the proof of Theorem 3.1, the proof of Theorem 3.2 is omitted here for brevity.

4 Numerical experiments

For both the PCCD scheme and its fast variant (FPCCD), the maximum discrete H^1 -norm error across all temporal levels is defined as:

$$E(M, N) = \max_{0 \leq j \leq N} \|e^j\|_{H^1} \quad \text{and} \quad \bar{E}(M, N) = \max_{0 \leq j \leq N} \|\bar{e}^j\|_{H^1},$$

where M and N represent the number of spatial and temporal mesh intervals, respectively. The temporal and spatial convergence orders are computed by

$$Order_t = \log_2 \left(\frac{E(M, N)}{E(M, 2N)} \right), \quad Order_s = \log_2 \left(\frac{E(M, N)}{E(2M, N)} \right).$$

Example 4.1 [Smooth Solution [22]], We first consider time-fractional Benjamin–Bona–Mahony–Burgers equation:

$$\begin{cases} {}^C D_0^\alpha (u - u_{xx}) + uu_x - u_{xxx} = f(x, t, u), & (x, t) \in \Omega \times (0, T], \\ u(x, 0) = 0, & x \in \bar{\Omega} \\ u(x, t) = 0, & u(x, t) \in \partial\Omega \times (0, T] \end{cases}$$

where $\Omega = [0, 1]$, with the forcing term

$$f(x, t, u) = (1 + \pi^2) \sin(\pi x) + \pi^2 \cdot u + \frac{\pi \cos(\pi x)}{\Gamma(1 + \alpha)t^\alpha} \cdot u.$$

The exact solution $u(x, t) = \frac{\sin(\pi x)}{\Gamma(1 + \alpha)} t^\alpha$, which exhibits smooth regularity. Unless otherwise stated, all computations in this example are performed for the time interval $T = 1$ with the grading parameter $r = 3/(2\alpha)$.

Table 1 reports the temporal accuracy of the PCCD scheme for $\alpha = 0.4, 0.6, 0.8$. We set the number of time steps N equal to the spatial grid number M and increase them as 6, 12, 24, 48, 96. The observed temporal convergence order approaches 2, which agrees with our theoretical analysis.

Table 1 Temporal accuracy of PCCD scheme for example 4.1 ($M = N$)

N	$\alpha = 0.4$		$\alpha = 0.6$		$\alpha = 0.8$	
	$E(M, N)$	$Order_t$	$E(M, N)$	$Order_t$	$E(M, N)$	$Order_t$
6	1.295e-01	—	1.285e-01	—	1.233e-01	—
12	3.073e-02	2.075	3.051e-02	2.074	2.927e-02	2.075
24	7.435e-03	2.048	7.380e-03	2.047	7.080e-03	2.048
48	1.823e-03	2.027	1.811e-03	2.027	1.737e-03	2.026
96	4.514e-04	2.014	4.483e-04	2.014	4.300e-04	2.014

Table 2 Space accuracy of PCCD scheme for example 4.1 ($N = 2000$)

M	$\alpha = 0.4$		$\alpha = 0.6$		$\alpha = 0.8$	
	$E(M, N)$	$Order_s$	$E(M, N)$	$Order_s$	$E(M, N)$	$Order_s$
4	6.367e-03	—	5.813e-03	—	4.907e-03	—
8	4.340e-04	3.875	3.958e-04	3.876	3.342e-04	3.876
16	2.791e-05	3.959	2.542e-05	3.961	2.136e-05	3.968
32	1.786e-06	3.966	1.607e-06	3.984	1.343e-06	3.992
64	1.157e-07	3.948	1.016e-07	3.984	8.419e-08	3.996

Table 3 Comparison between PCCD and FPCCD schemes for example 4.1 ($\alpha = 0.5$, $N = 20000$)

M	PCCD			FPCCD		
	$E(M, N)$	$Order_s$	CPU time (s)	$E(M, N)$	$Order_s$	CPU time (s)
4	6.367e-03	—	429.884	6.372e-03	—	5.531
8	4.339e-04	3.875	411.312	4.345e-04	3.874	5.375
16	2.787e-05	3.960	475.167	2.789e-05	3.962	5.125
32	1.769e-06	3.978	464.611	1.767e-06	3.980	5.359
64	1.115e-07	3.989	479.283	1.088e-07	4.022	6.500

Table 4 Comparison between PCCD scheme and scheme in [27] for example 4.1 ($\alpha = 0.5$, $N = 2000$)

M	PCCD			Scheme in [27]		
	$E(M, N)$	$Order_s$	CPU time (s)	$E(M, N)$	$Order_s$	CPU time (s)
4	6.153e-03	—	3.093	1.016e-01	—	3.109
8	4.219e-04	3.875	3.093	2.850e-02	1.834	3.063
16	2.714e-05	3.960	2.969	7.579e-03	1.911	3.219
32	1.714e-06	3.978	3.313	1.952e-03	1.957	3.406
64	1.091e-07	3.989	4.123	4.952e-04	1.979	3.422

Table 2 shows the spatial accuracy for the same α values. Here we fix $N = 2000$ and double the spatial grids $M = 4, 8, 16, 32, 64$. The H^1 -norm convergence order tends to 4, confirming the expected fourth-order spatial accuracy.

Table 3 compares the computational efficiency of the PCCD and FPCCD schemes for $\alpha = 0.5$ with $N = 20000$ and $M = 4, 8, 16, 32, 64$. Under similar error levels $E(M, N)$, the FPCCD scheme saves significant CPU time, consistent with our theoretical expectations.

Table 4 compares the PCCD scheme with the second-order method of Zhou et al. [27] for $\alpha = 0.5$ ($N = 2000$). While both schemes exhibit comparable computational costs, the PCCD scheme achieves markedly higher accuracy (fourth-order spatial convergence) versus the second-order convergence of the reference scheme.

The long-time behavior of the scheme is illustrated in Fig. 1, where the numerical solution is computed up to $T = 10$ with $\alpha = 0.5$. The left panel displays the top view of the solution, and the right panel shows cross-sectional profiles at $t = 2, 4, 6, 8, 10$. The results demonstrate that the proposed finite difference scheme maintains high accuracy even in long-time simulations.

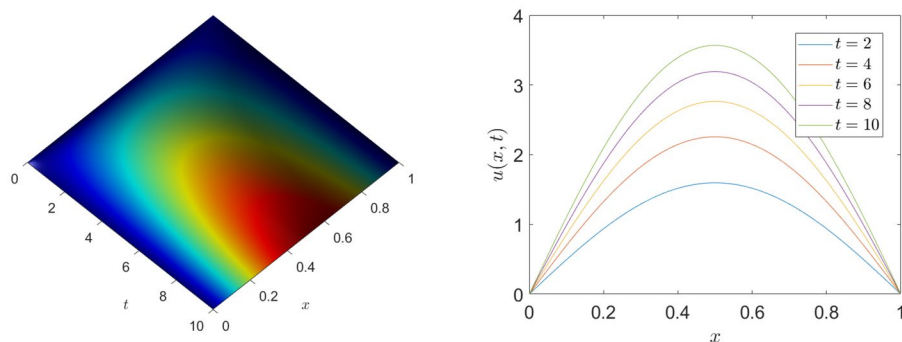


Fig. 1 Long-time simulation up to $T = 10$: left panel shows the top view of the numerical solution; right panel shows solution profiles at $t = 2, 4, 6, 8, 10$ with $\alpha = 0.5$

Example 4.2 [Low-Regularity Initial Data [26]], To stress-test the scheme under non-ideal conditions, we consider the time-fractional Benjamin–Bona–Mahony–Burgers equation:

$$\begin{cases} {}^C D_0^\alpha (u - u_{xx}) + uu_x - u_{xx} = f(x, t, u), & (x, t) \in \Omega \times (0, T], \\ u(x, 0) = \varphi(x), & x \in \bar{\Omega}, \\ u(x, t) = 0, & (x, t) \in \partial\Omega \times (0, T], \end{cases}$$

where $\Omega = [0, 1]$, with the initial function $\varphi(x) = \sin^{\frac{5}{2}}(\pi x)$, and the forcing term

$$f(x, t, u) = \varphi(x) - \left[2 + \frac{t^\alpha}{\Gamma(1 + \alpha)} \right] \varphi''(x) + u \cdot \left[1 + \frac{t^\alpha}{\Gamma(1 + \alpha)} \right] \varphi'(x).$$

The exact solution is

$$u(x, t) = \left[1 + \frac{t^\alpha}{\Gamma(1 + \alpha)} \right] \varphi(x).$$

Clearly, it exhibits lower spatial regularity because $\varphi(x) \in C^2([0, 1])$ but $\varphi(x) \notin C^3([0, 1])$.

As in Example 4.1, the computations in this example are performed for $T = 1$ with the grading parameter $r = 3/(2\alpha)$ unless otherwise stated.

Table 5 reports the temporal convergence of the PCCD scheme for $\alpha = 0.4, 0.6, 0.8$. The observed order $Order_t \approx 2$ across all α demonstrates that the time-stepping mechanism remains robust even with irregular initial data.

Table 6 shows the spatial convergence of the PCCD scheme. Due to the lack of C^3 -regularity, the convergence rate drops to about 2, deviating from the theoretical fourth-order behavior. This confirms that the compact difference framework requires sufficient solution smoothness to preserve high-order spatial accuracy, in agreement with findings in [26].

Table 5 Temporal accuracy of PCCD scheme for example 4.2 ($M = N$)

N	$\alpha = 0.4$		$\alpha = 0.6$		$\alpha = 0.8$	
	$E(M, N)$	$Order_t$	$E(M, N)$	$Order_t$	$E(M, N)$	$Order_t$
6	9.715e-02	—	9.576e-02	—	9.391e-02	—
12	2.248e-02	2.111	2.239e-02	2.096	2.200e-02	2.094
24	5.831e-03	1.947	5.813e-03	1.946	5.699e-03	1.948
48	1.478e-03	1.980	1.473e-03	1.981	1.443e-03	1.982
96	3.711e-04	1.994	3.697e-04	1.994	3.619e-04	1.995

Table 6 Space accuracy of PCCD scheme for example 4.2 ($N = 2000$)

M	$\alpha = 0.4$		$\alpha = 0.6$		$\alpha = 0.8$	
	$E(M, N)$	$Order_s$	$E(M, N)$	$Order_s$	$E(M, N)$	$Order_s$
8	4.946e-02	—	4.942e-02	—	4.859e-02	—
16	1.286e-02	1.944	1.282e-02	1.946	1.258e-02	1.950
32	3.304e-03	1.960	3.293e-03	1.961	3.226e-03	1.963
64	8.332e-04	1.987	8.303e-04	1.988	8.128e-04	1.989
128	2.089e-04	1.996	2.081e-04	1.996	2.037e-04	1.997

Table 7 Comparison between PCCD scheme and scheme in [27] for example 4.2 ($\alpha = 0.5$, $N = 2000$)

M	PCCD			Scheme in [27]		
	$E(M, N)$	$Order_s$	CPU time (s)	$E(M, N)$	$Order_s$	CPU time (s)
8	4.955e-02	—	3.250	2.935e-01	—	3.046
16	1.287e-02	1.944	3.234	7.409e-02	1.986	3.046
32	3.307e-03	1.960	3.250	2.038e-02	1.862	3.266
64	8.338e-04	1.988	3.609	5.391e-03	1.919	3.328
128	2.090e-04	1.996	10.938	1.415e-03	1.930	10.938

Table 8 Comparison between FPCCD scheme and scheme in [27] for example 4.2 ($\alpha = 0.5$, $N = 2000$)

M	FPCCD			Scheme in [27]		
	$E(M, N)$	$Order_s$	CPU time (s)	$E(M, N)$	$Order_s$	CPU time (s)
8	4.957e-02	—	1.062	2.935e-01	—	3.046
16	1.287e-02	1.945	0.718	7.409e-02	1.986	3.046
32	3.308e-03	1.960	0.468	2.038e-02	1.862	3.266
64	8.340e-04	1.988	0.671	5.391e-03	1.919	3.328
128	2.091e-04	1.996	3.812	1.415e-03	1.930	10.938

Tables 7 and 8 compare the PCCD and FPCCD schemes with the second-order method of Zhou et al. [27]. Under identical parameter configurations ($\alpha = 0.5$, $N = 2000$), both PCCD and FPCCD achieve nearly an order of magnitude higher precision than the reference scheme, despite the reduced spatial order due to limited regularity. Moreover, the FPCCD scheme significantly reduces computational time, as shown in Table 8.

The long-time behavior of the scheme is illustrated in Fig. 2, where the numerical solution is computed up to $T = 10$ with $\alpha = 0.5$. The left panel displays the top view of the solution, and the right panel shows cross-sectional profiles at $t = 2, 4, 6, 8, 10$.

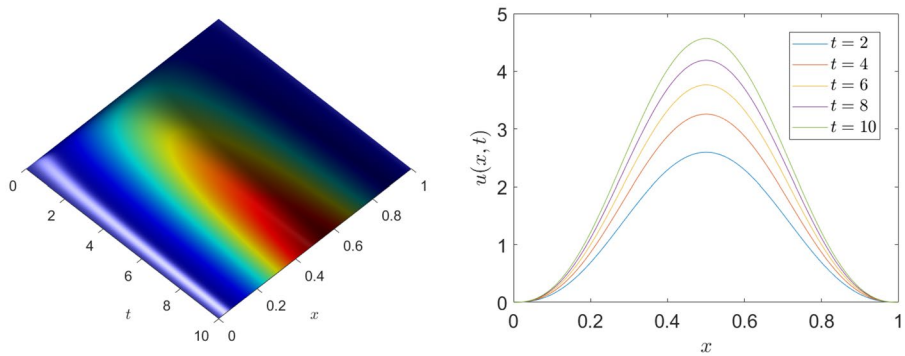


Fig. 2 Long-time simulation up to $T = 10$: left panel shows the top view of the numerical solution; right panel shows solution profiles at $t = 2, 4, 6, 8, 10$ with $\alpha = 0.5$

The results further confirm that the proposed scheme maintains stability and accuracy in long-time simulations.

5 Conclusion

This work develops a numerical method for nonlinear time-fractional Benjamin-Bona-Mahony equations by integrating predictor-corrector schemes, order-reduction techniques, and two compact difference operators. The proposed scheme achieves fourth-order spatial accuracy and second-order temporal accuracy. Discrete energy methods are employed to analyze the scheme, yielding error estimates in the H^1 -norm. To optimize computational efficiency, exponential sum approximations are implemented, enabling a fast solution algorithm that significantly reduces computational costs. Numerical experiments validate the effectiveness of our methods.

Acknowledgements We sincerely thank the anonymous reviewers and the associate editor for their thoughtful comments and constructive suggestions, which have significantly improved the quality of the manuscript. This work was supported by the Sichuan Science and Technology Program (Grant No. 2024NSFSC0441).

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Benjamin, T.B., Bona, J.L., Mahony, J.J.: Model equations for long waves in nonlinear dispersive systems. *Philos. Trans. R. Soc. Lond. A.* **272**(1220), 47–78 (1972)
2. Zhang, Q., Liu, L.: Convergence and stability in maximum norms of linearized fourth-order conservative compact scheme for benjamin–bona–mahony–burgers’ equation. *J. Sci. Comput.* **87**(2), 59 (2021)

3. Zhang, Q., Liu, L., Zhang, J.: The numerical analysis of two linearized difference schemes for the benjamin–bona–mahony–burgers equation. *Numer. Methods. Partial. Differ. Equations.* **36**(6), 1790–1810 (2020)
4. Dehghan, M., Abbaszadeh, M., Mohebbi, A.: The numerical solution of nonlinear high dimensional generalized benjamin–bona–mahony–burgers equation via the meshless method of radial basis functions. *Comput. Math. Appl.* **68**(3), 212–237 (2014)
5. Dehghan, M., Abbaszadeh, M., Mohebbi, A.: The use of interpolating element-free galerkin technique for solving 2d generalized benjamin–bona–mahony–burgers and regularized long-wave equations on non-rectangular domains with error estimate. *J. Comput. Appl. Math.* **286**, 211–231 (2015)
6. Omrani, K., Ayadi, M.: Finite difference discretization of the benjamin–bonamahony–burgers equation. *Numer. Methods Partial Differ. Equations: Int. J.* **24**(1), 239–248 (2008)
7. Dehghan, M., Abbaszadeh, M., Deng, W.: Fourth-order numerical method for the space–time tempered fractional diffusion-wave equation. *Appl. Math. Lett.* **73**, 120–127 (2017)
8. Dehghan, M., Abbaszadeh, M.: Analysis of the element free galerkin (efg) method for solving fractional cable equation with dirichlet boundary condition. *Appl. Numer. Math.* **109**, 208–234 (2016)
9. Saadatmandi, A., Dehghan, M.: A new operational matrix for solving fractional order differential equations. *Comput. Math. Appl.* **59**(3), 1326–1336 (2010)
10. Ran, M., Zhou, X.: An implicit difference scheme for the time-fractional cahnhilliard equations. *Math. Comput. Simul.* **180**, 61–71 (2021)
11. Guner, O., Bekir, A.: Bright and dark soliton solutions for some nonlinear fractional differential equations. *Chin. Phys. B* **25**(3), 030203 (2016)
12. Song, L., Zhang, H.: Solving the fractional bbm–burgers equation using the homotopy analysis method. *Chaos, Solitons Fractals* **40**(4), 1616–1622 (2009)
13. Ahmad, J., Masood, K., Ayub, F., Shah, N.A.: Bifurcation analysis and novel wave patterns to zakharov–kuznetsov–benjamin–bona–mahony equation with truncated m-fractional derivative. *Sci. Rep.* **15**(1), 35269 (2025)
14. Zhang, W., Wang, H., Zhang, H., Li, Z., Li, X.: Dynamical behavior of the fractional bbmb equation on unbounded domain. *Fractal Fract.* **8**(7), 383 (2024)
15. Lyu, P., Vong, S.: A linearized second-order finite difference scheme for time fractional generalized bbm equation. *Appl. Math. Lett.* **78**, 16–23 (2018)
16. Li, C.: Linearized difference schemes for a bbm equation with a fractional nonlocal viscous term. *Appl. Math. Comput.* **311**, 240–250 (2017)
17. Yi, Y.-G., Xie, B., Wang Y.-M.: A high-order implicit–explicit compact finite difference method for a nonlinear time-fractional benjamin–bona–mahony–burgers equation. *J. Math. Chem.* **64**(2), 16 (2026)
18. Lin, J., Shi, L., Reutskiy, S., Lu, J.: Numerical treatment of multi-dimensional time-fractional benjamin–bona–mahony–burgers equations in arbitrary domains with a novel improvised rbf-based method. *Comput. Math. Appl.* **167**, 178–198 (2024)
19. Dhayal, R.: Successive approximation and optimal controls for stochastic benjamin–bona–mahony equation. *J. Appl. Math. Comput.* 1–19 (2025)
20. Dhayal, R., Malik, M., Abbas, S., Debbouche, A.: Optimal controls for second order stochastic differential equations driven by mixed-fractional brownian motion with impulses. *Math. Methods Appl. Sci.* **43**(7), 4107–4124 (2020)
21. Dhayal, R., Malik, M., Abbas, S.: Solvability and optimal controls of noninstantaneous impulsive stochastic fractional differential equation of order $q \in (1, 2)$. *Stochastics*. **93**(5), 780–802 (2021)
22. Lyu, P., Vong, S.: A high-order method with a temporal nonuniform mesh for a time-fractional benjamin–bona–mahony equation. *J. Sci. Comput.* **80**(3), 1607–1628 (2019)
23. McLean, W.: Regularity of solutions to a time-fractional diffusion equation. *Anziam J.* **52**(2), 123–138 (2010)
24. Stynes, M., O’Riordan, E., Gracia, J.L.: Error analysis of a finite difference method on graded meshes for a time-fractional diffusion equation. *SIAM J. Numer. Anal.* **55**(2), 1057–1079 (2017)
25. Chen, H., Stynes, M.: Error analysis of a second-order method on fitted meshes for a time-fractional diffusion problem. *J. Sci. Comput.* **79**, 624–647 (2019)
26. Wang, Y.-M.: A high-order linearized and compact difference method for the time-fractional benjamin–bona–mahony equation. *Appl. Math. Lett.* **105**, 106339 (2020)
27. Zhou, Y., Li, C., Stynes M.: A fast second-order predictor–corrector method for a nonlinear time-fractional benjamin–bona–mahony–burgers equation. *Numer. Algorithms* **95**(2), 693–720 (2024)

28. Nguyen, T.B., Jang, B.: A high-order predictor-corrector method for solving nonlinear differential equations of fractional order. *Fract. Calc. Appl. Anal.* **20**(2), 447–476 (2017)
29. Wang, X., Zhang, Q., Sun Z.-Z.: The pointwise error estimates of two energypreserving fourth-order compact schemes for viscous burgers' equation. *Adv. Comput. Math.* **47**, 1–42 (2021)
30. Jiang, S., Zhang, J., Zhang, Q., Zhang, Z.: Fast evaluation of the caputo fractional derivative and its applications to fractional diffusion equations. *Commun. Comput. Phys.* **21**(3), 650–678 (2017)
31. Diethelm, K., Ford, N.: The analysis of fractional differential equations. *Lect. Notes Math.* 2004 (2010)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.